Bounded-Arboricity Subcases of Two Random-Matrix Norm Conjectures

Candidate partial result for arXiv:2502.02186

9 August 2026

Status. Candidate partial result, likely valid, pending human review. The packet proves the sharp-parameter Gaussian upper conjecture for variance profiles whose bipartite support graph has bounded arboricity. It proves the Bernoulli conjecture on the same class; in particular, both statements hold with absolute constants on forest supports. It also proves the Bernoulli conjecture at $p = 1$ or $q = \infty$ for arbitrary support. The unrestricted conjectures remain open here.

1 Source and exact questions

The source is R. Latała and M. Strzelecka, *Operator $\ell_p \to \ell_q$ norms of Gaussian matrices*, arXiv:2502.02186v3 [1]. On page 3, Conjecture 5 asks whether, for every $1 \le p \le 2 \le q \le \infty$ and every deterministic $m \times n$ matrix $A = (a_{ij})$,

$$\mathbb{E}\left\|(a_{ij}g_{ij})\right\|_{p\to q} \lesssim \sqrt{p^* \wedge \operatorname{Log} n}\,\max_i \left\|(a_{ij})_j\right\|_{p^*} + \sqrt{q \wedge \operatorname{Log} m}\,\max_j \left\|(a_{ij})_i\right\|_q + \mathbb{E}\max_{i,j}|a_{ij}g_{ij}|, \quad (1)$$

with an absolute implicit constant. Here p^* is the Holder conjugate and $\operatorname{Log} x = 1 \vee \log x$ for $x > 0$.

On page 5, equation (3) conjectures a Bernoulli analogue. To state it unambiguously, put

$$\Phi_{p,q}(A) := \max_{0\le k\le mn}\ \inf_{\substack{I\subseteq[m]\times[n]\\|I|=k}}\ \sup_{\substack{\|s\|_{q^*}\le 1\\ \|t\|_p\le 1}} \left\|\sum_{(i,j)\notin I} a_{ij}\varepsilon_{ij}s_it_j\right\|_{\operatorname{Log} k}, \quad (2)$$

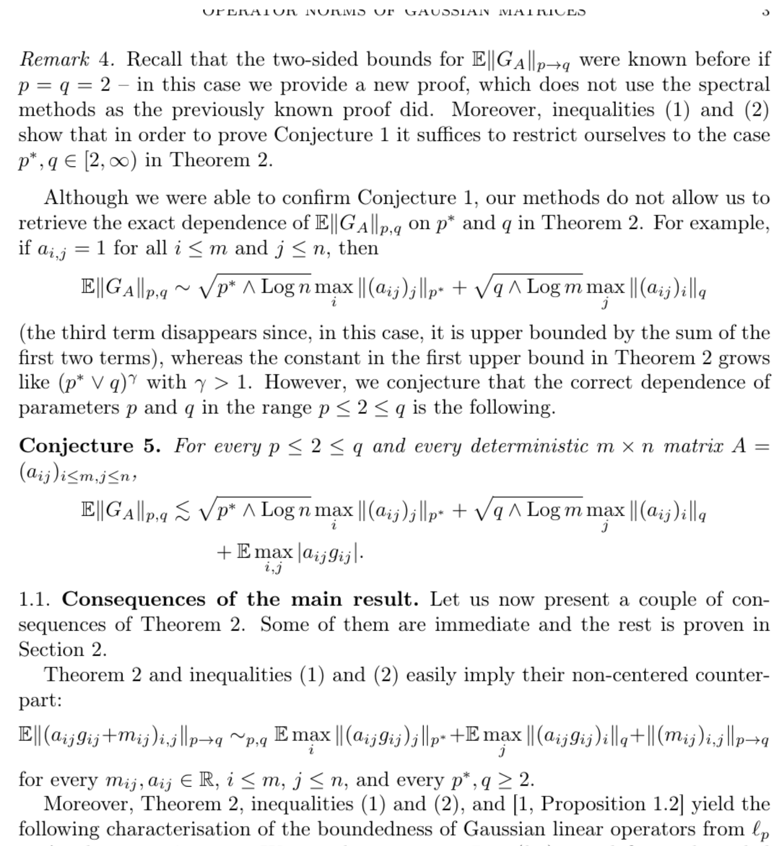
where $\operatorname{Log} 0 = 1$. The question is whether

$$\mathbb{E}\left\|(a_{ij}\varepsilon_{ij})\right\|_{p\to q} \asymp_{p,q} \max_i \left\|(a_{ij})_j\right\|_{p^*} + \max_j \left\|(a_{ij})_i\right\|_q + \Phi_{p,q}(A). \quad (3)$$

2 Definitions and notation

The *bipartite support graph* $\Gamma(A)$ has row vertices $[m]$, column vertices $[n]$, and an edge (i, j) exactly when $a_{ij} \neq 0$. Its arboricity $\mathfrak{a}(A)$ is the least number of forests into which its edge set can be partitioned. Thus $\mathfrak{a}(A) = 1$ when the support is a nonempty forest. Set

$$R_{p^*}(A) = \max_i \left\|(a_{ij})_j\right\|_{p^*}, \qquad C_q(A) = \max_j \left\|(a_{ij})_i\right\|_q, \qquad M(A) = \mathbb{E}\max_{i,j}|a_{ij}g_{ij}|.$$



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Remark 4. Recall that the two-sided bounds for $\mathbb{E}\|G_A\|_{p\to q}$ were known before if $p = q = 2$ – in this case we provide a new proof, which does not use the spectral methods as the previously known proof did. Moreover, inequalities (1) and (2) show that in order to prove Conjecture 1 it suffices to restrict ourselves to the case $p^*, q \in [2,\infty)$ in Theorem 2.

Although we were able to confirm Conjecture 1, our methods do not allow us to retrieve the exact dependence of $\mathbb{E}\|G_A\|_{p,q}$ on p^* and q in Theorem 2. For example, if $a_{i,j} = 1$ for all $i \le m$ and $j \le n$, then

$$\mathbb{E}\|G_A\|_{p,q} \sim \sqrt{p^* \wedge \operatorname{Log} n}\max_i \|(a_{ij})_j\|_{p^*} + \sqrt{q \wedge \operatorname{Log} m}\max_j \|(a_{ij})_i\|_q$$

(the third term disappears since, in this case, it is upper bounded by the sum of the first two terms), whereas the constant in the first upper bound in Theorem 2 grows like $(p^* \vee q)^\gamma$ with $\gamma > 1$. However, we conjecture that the correct dependence of parameters p and q in the range $p \le 2 \le q$ is the following.

Conjecture 5. *For every $p \le 2 \le q$ and every deterministic $m \times n$ matrix $A = (a_{ij})_{i\le m, j\le n}$,*

$$\mathbb{E}\|G_A\|_{p,q} \lesssim \sqrt{p^* \wedge \operatorname{Log} n}\max_i \|(a_{ij})_j\|_{p^*} + \sqrt{q \wedge \operatorname{Log} m}\max_j \|(a_{ij})_i\|_q + \mathbb{E}\max_{i,j}|a_{ij}g_{ij}|.$$

1.1. **Consequences of the main result.** Let us now present a couple of consequences of Theorem 2. Some of them are immediate and the rest is proven in Section 2.

Theorem 2 and inequalities (1) and (2) easily imply their non-centered counterpart:

$$\mathbb{E}\|(a_{ij}g_{ij}+m_{ij})_{i,j}\|_{p\to q} \sim_{p,q} \mathbb{E}\max_i \|(a_{ij}g_{ij})_j\|_{p^*} + \mathbb{E}\max_j \|(a_{ij}g_{ij})_i\|_q + \|(m_{ij})_{i,j}\|_{p\to q}$$

for every $m_{ij}, a_{ij} \in \mathbb{R}$, $i \le m$, $j \le n$, and every $p^*, q \ge 2$.

Moreover, Theorem 2, inequalities (1) and (2), and [1, Proposition 1.2] yield the following characterisation of the boundedness of Gaussian linear operators from ℓ_p

Figure 1: The sharp-parameter Gaussian conjecture, source page 3.

3 Proof intuition

A forest can be rooted component by component. Assign each edge to its child and split the edges according to whether the child is a row vertex or a column vertex. In the first piece every row has degree at most one; in the second piece every column has degree at most one. When $p \le q$, the corresponding operator norms are exactly the largest column ℓ_q norm and the largest row ℓ_{p^*} norm. This gives a deterministic row-plus-column estimate on a forest, and an arboricity multiple of that estimate in general.

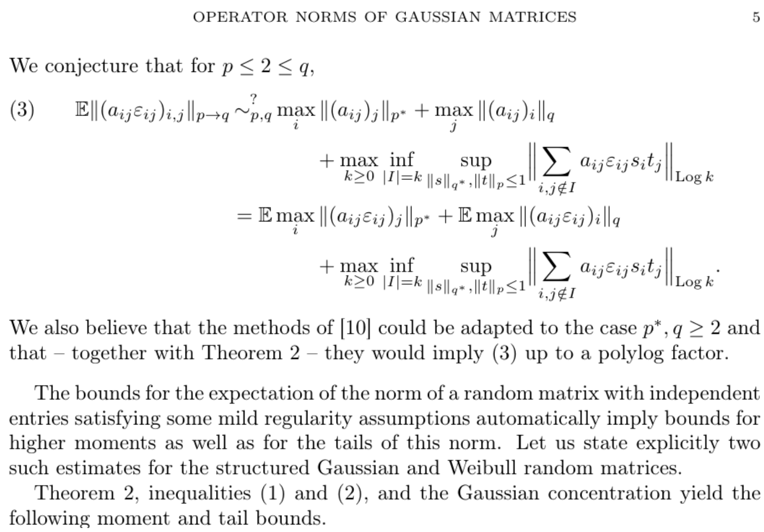
For Gaussian entries, the deterministic split reduces the conjecture to the maximal row and column estimates already established as equations (1) and (2) of the source. The elementary bound $\|x\|_r \le d^{1/r}\|x\|_\infty$ when $r \ge \operatorname{Log} d$ supplies the dimension truncations in Conjecture 5. For Bernoulli entries, all row and column norms are deterministic. The same forest estimate also bounds every deleted-edge chaos appearing in $\Phi_{p,q}(A)$.

4 Formal results

Lemma 1 (Deterministic arboricity estimate)**.** *Let $1 \le p \le q \le \infty$, and let $B = (b_{ij})$ be nonzero. If $a = \mathfrak{a}(B)$, then*

$$\max\{R_{p^*}(B), C_q(B)\} \le \|B\|_{p\to q} \le a\big(R_{p^*}(B) + C_q(B)\big). \tag{4}$$

For a forest-supported matrix, the last factor a is 1.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We conjecture that for $p \le 2 \le q$,

$$\mathbb{E}\|(a_{ij}\varepsilon_{ij})_{i,j}\|_{p\to q} \overset{?}{\sim}_{p,q} \max_i \|(a_{ij})_j\|_{p^*} + \max_j \|(a_{ij})_i\|_q + \max_{k\ge 0} \inf_{|I|=k} \sup_{\|s\|_{q^*},\|t\|_p\le 1} \Big\| \sum_{i,j\notin I} a_{ij}\varepsilon_{ij}s_it_j \Big\|_{\mathrm{Log}\,k} = \mathbb{E}\max_i \|(a_{ij}\varepsilon_{ij})_j\|_{p^*} + \mathbb{E}\max_j \|(a_{ij}\varepsilon_{ij})_i\|_q + \max_{k\ge 0} \inf_{|I|=k} \sup_{\|s\|_{q^*},\|t\|_p\le 1} \Big\| \sum_{i,j\notin I} a_{ij}\varepsilon_{ij}s_it_j \Big\|_{\mathrm{Log}\,k}. \tag{3}$$

We also believe that the methods of [10] could be adapted to the case $p^*, q \ge 2$ and that – together with Theorem 2 – they would imply (3) up to a polylog factor.

The bounds for the expectation of the norm of a random matrix with independent entries satisfying some mild regularity assumptions automatically imply bounds for higher moments as well as for the tails of this norm. Let us state explicitly two such estimates for the structured Gaussian and Weibull random matrices.

Theorem 2, inequalities (1) and (2), and the Gaussian concentration yield the following moment and tail bounds.

Figure 2: The Bernoulli conjecture, source page 5, equation (3).

Proof. The lower bounds follow by testing B on a coordinate vector for the column bound, and by testing one row as a functional on ℓ_p^n for the row bound.

First suppose that $\Gamma(B)$ is a forest. Root each nontrivial component. Partition its edges as $E_r \sqcup E_c$, according as the child endpoint is a row or a column vertex, and write $B = B_r + B_c$ for the corresponding matrices. Every row of B_r has at most one nonzero entry. If that entry in row i lies in column $j(i)$, then for $x \in \mathbb{R}^n$,

$$\|B_r x\|_q^q = \sum_j |x_j|^q \sum_{i:j(i)=j} |b_{ij}|^q \le C_q(B_r)^q \, \|x\|_q^q \le C_q(B_r)^q \, \|x\|_p^q \, ,$$

with the usual modification when $q = \infty$. Equality is obtained by a coordinate vector, so $\|B_r\|_{p\to q} = C_q(B_r)$.

Every column of B_c has at most one nonzero entry. Grouping columns by their incident row and applying Holder's inequality gives

$$|(B_c x)_i| \le \left\|(b_{ij})_{j:(i,j)\in E_c}\right\|_{p^*} \left\|(x_j)_{j:(i,j)\in E_c}\right\|_p .$$

The groups are disjoint, and $p \le q$, hence

$$\|B_c x\|_q \le R_{p^*}(B_c) \left(\sum_i \left\|(x_j)_{j:(i,j)\in E_c}\right\|_p^p \right)^{1/p} = R_{p^*}(B_c) \, \|x\|_p \, .$$

Again equality follows by concentrating on one row, so $\|B_c\|_{p\to q} = R_{p^*}(B_c)$. The triangle inequality now gives

$$\|B\|_{p\to q} \le C_q(B_r) + R_{p^*}(B_c) \le C_q(B) + R_{p^*}(B).$$

The same calculation uses maxima instead of sums at an infinite endpoint.

For general arboricity a, partition the support into a forests, restrict B to those edge sets, and apply the forest estimate plus the triangle inequality. Each restricted row and column norm is no larger than the corresponding norm of B, proving (4). $\square$

Lemma 2 (Dimension-truncated Gaussian row maxima)**.** *For $2 \le r \le \infty$ and every deterministic $u \times d$ matrix $D = (d_{ij})$,*

$$\mathbb{E} \max_i \|(d_{ij} g_{ij})_j\|_r \le C \left(\sqrt{r \wedge \operatorname{Log} d} \max_i \|(d_{ij})_j\|_r + \mathbb{E} \max_{i,j} |d_{ij} g_{ij}| \right) \tag{5}$$

for an absolute constant C. The analogous column statement also holds.

Proof. Equation (2) of [1] gives the right side of (5) with $\sqrt{r}$ in place of $\sqrt{r \wedge \operatorname{Log} d}$. This proves the claim when $r \le \operatorname{Log} d$. If $r \ge \operatorname{Log} d$, then pointwise

$$\max_i \|(d_{ij} g_{ij})_j\|_r \le d^{1/r} \max_{i,j} |d_{ij} g_{ij}| \le e \max_{i,j} |d_{ij} g_{ij}|.$$

This also covers $r = \infty$. Transposition and equation (1) of [1] give the column statement. □

Theorem 3 (Gaussian conjecture for bounded arboricity)**.** *Let $1 \le p \le 2 \le q \le \infty$, and let A have nonempty support of arboricity a. Then*

$$\mathbb{E} \|(a_{ij} g_{ij})\|_{p \to q} \le Ca \Big(\sqrt{p^* \wedge \operatorname{Log} n}\, R_{p^*}(A) + \sqrt{q \wedge \operatorname{Log} m}\, C_q(A) + M(A) \Big), \tag{6}$$

where C is absolute. Consequently Conjecture 5 holds with an absolute constant for every forest-supported variance profile.

Proof. Apply Lemma 1 pointwise to the random matrix $G_A = (a_{ij} g_{ij})$. Its support is almost surely the support of A, so

$$\|G_A\|_{p \to q} \le a \left(\max_i \|(a_{ij} g_{ij})_j\|_{p^*} + \max_j \|(a_{ij} g_{ij})_i\|_q \right).$$

Take expectations and apply Lemma 2 to rows and columns. The two copies of $M(A)$ are absorbed into the absolute constant. □

Theorem 4 (Bernoulli conjecture for bounded arboricity)**.** *Let $1 \le p \le q \le \infty$, let A have nonempty support of arboricity a, and put*

$$H_{p,q}(A) = R_{p^*}(A) + C_q(A) + \Phi_{p,q}(A).$$

Then

$$\frac{1}{2(a+1)} H_{p,q}(A) \le \mathbb{E} \|(a_{ij} \varepsilon_{ij})\|_{p \to q} \le a H_{p,q}(A). \tag{7}$$

In particular, the Bernoulli conjecture (3) *holds with absolute constants on forest supports, throughout the source range $p \le 2 \le q$ (indeed, throughout $p \le q$).*

Proof. Bernoulli signs do not change row or column norms. Lemma 1 gives, for every realization,

$$\max\{R_{p^*}(A), C_q(A)\} \le \|(a_{ij} \varepsilon_{ij})\|_{p \to q} \le a(R_{p^*}(A) + C_q(A)).$$

Every matrix obtained by deleting entries of A has arboricity at most a. For fixed I, s, t and every sign realization, Lemma 1 therefore gives

$$\left| \sum_{(i,j) \notin I} a_{ij} \varepsilon_{ij} s_i t_j \right| \le a(R_{p^*}(A) + C_q(A)).$$

Taking the indicated scalar $L_{\operatorname{Log} k}$ norm, supremum, infimum, and maximum shows $\Phi_{p,q}(A) \le a(R_{p^*}(A) + C_q(A))$. Thus

$$H_{p,q}(A) \le (a+1)(R_{p^*}(A) + C_q(A)) \le 2(a+1)\mathbb{E}\left\|(a_{ij}\varepsilon_{ij})\right\|_{p\to q},$$

which is the left inequality in (7). The pointwise upper bound also gives $\mathbb{E}\left\|(a_{ij}\varepsilon_{ij})\right\|_{p\to q} \le a(R_{p^*}(A)+C_q(A)) \le aH_{p,q}(A)$. $\square$

Corollary 5 (Arbitrary-support Bernoulli endpoints)**.** *For arbitrary A and $1 \le q \le \infty$, equation* (3) *holds at $p = 1$, with*

$$\mathbb{E}\left\|(a_{ij}\varepsilon_{ij})\right\|_{1\to q} = C_q(A), \qquad C_q(A) \le H_{1,q}(A) \le 3C_q(A).$$

For arbitrary A and $1 \le p \le \infty$, it holds at $q = \infty$, with

$$\mathbb{E}\left\|(a_{ij}\varepsilon_{ij})\right\|_{p\to\infty} = R_{p^*}(A), \qquad R_{p^*}(A) \le H_{p,\infty}(A) \le 3R_{p^*}(A).$$

Proof. The $\ell_1 \to \ell_q$ norm is the largest column ℓ_q norm, hence is $C_q(A)$ for every sign realization. Also $R_\infty(A) \le C_q(A)$, and each deleted-edge matrix has $\ell_1 \to \ell_q$ norm at most $C_q(A)$. Therefore $\Phi_{1,q}(A) \le C_q(A)$, giving the first assertion. The second is the dual argument: the $\ell_p \to \ell_\infty$ norm is the largest row ℓ_{p^*} norm, $C_\infty(A) \le R_{p^*}(A)$, and deleting entries cannot increase this norm. $\square$

5 Verification notes

- The rooted split was audited at both possible root parities. An edge is assigned to its unique child, so the row-child piece has row degree at most one and the column-child piece has column degree at most one.
- The only exponent relation used in Lemma 1 is $\|z\|_q \le \|z\|_p$ for $p \le q$; this includes all source parameters $p \le 2 \le q$ and the endpoints.
- Deleting edges cannot increase arboricity, row norms, or column norms. This is exactly what is needed for the infimum over I in $\Phi_{p,q}(A)$.
- The Gaussian argument uses only equations (1) and (2) already proved in the source, plus the pointwise ℓ_r-to-ℓ_∞ comparison. No unproved probabilistic assertion is imported.
- The script `code/verify_forest_bounds.py` tested 500 random bipartite forests and 10,000 Bernoulli signings. It checked the rooted decomposition and the exact spectral/endpoint inequalities. The command was

  ```
  conda run --no-capture-output -n sandbox python
  code/verify_forest_bounds.py
  ```

  and it returned `PASS`. These finite tests are sanity checks, not part of the proof.

6 Limitations and novelty check

Theorems 3 and 4 have a factor linear in support arboricity. They do not give a uniform bound for arbitrary dense supports, so they do not settle either unrestricted conjecture. Corollary 5 settles only the two Bernoulli boundary families.

A bounded search on 9 August 2026 covered the run registry, solution index, attempt index, and proof-gap index using arXiv id 2502.02186 and the core norm keywords. It also searched arXiv for the exact title together with “forest support”, “support graph”, “star forest”, “Bernoulli”, and “bounded arboricity”, and searched for later arXiv papers citing the exact identifier. The searches found the source paper and unrelated graph/random matrix papers, but no statement of the forest/arboricity subcases or the arbitrary-support Bernoulli endpoint observation. Novelty confidence is moderate: the rooted-forest norm lemma is elementary and may be folklore even though its application to these two explicit conjectures was not located.

7 Human-review recommendation

Recommended for expert review as a strong, fully proved partial result. The highest-value checks are: (i) the interpretation of the deletion set I in the source’s equation (3); (ii) the degree-one norm identities at $p = 1$ and $q = \infty$; and (iii) whether the forest/arboricity application has appeared under different sparse-matrix terminology. The logical proof itself has no conditional step.

References

[1] R. Latała and M. Strzelecka, *Operator $\ell_p \to \ell_q$ norms of Gaussian matrices*, arXiv:2502.02186v3, 2025; final arXiv revision 9 June 2026. `https://arxiv.org/abs/2502.02186`.